

Flatcar Installing

Flatcar runs on cloud providers, virtualization platforms, and bare metal. All platforms use Ignition or Butane config passed as userdata for provisioning.

Cloud Providers

Provider	Support
Amazon EC2	Official
Microsoft Azure	Official
Google Compute Engine	Official
Hetzner	Official
DigitalOcean	Official
OpenStack	Official
Brightbox	Official
Akamai/Linode	Official
STACKIT	Official
VMware	Community
Scaleway	Community
OracleCloud	Community
OVHcloud	Community

Ignition config is passed via the provider's **user data** / **custom data** field at instance creation.

Virtual Machines

Platform	Support
QEMU/KVM	Official
libvirt	Official
KubeVirt	Community
Hyper-V	Community
VirtualBox	Community
Vagrant	Community

Platform	Support
Proxmox VE	Community

KubeVirt

KubeVirt images are available as `qcow2` for both `amd64` and `arm64` from Alpha 3975.0.0 onward.

Download:

```
wget https://alpha.release.flatcar-linux.net/amd64-usr/3975.0.0/flatcar_production_kubevirt_image.qcow2
```

Upload to cluster via virtctl:

```
virtctl image-upload pvc flatcar-production-kubevirt-image --size=10Gi \
  --image-path=flatcar_production_kubevirt_image.qcow2 \
  --uploadproxy-url https://cdi-uploadproxy:31001 --insecure
```

Or via PVC with CDI annotation:

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: flatcar-amd64-3975
  annotations:
    cdi.kubevirt.io/storage.import.endpoint: "https://alpha.release.flatcar-linux.net/amd64-usr/3975.0.0/flatcar_production_kubevirt_image.qcow2"
spec:
  accessModes: [ReadWriteOnce]
  resources:
    requests:
      storage: 10Gi
  storageClassName: ceph-block
```

Provisioning: Supports both **cloud-init** and **Ignition** via `cloudInitConfigDrive`.

VM with Ignition userdata:

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
```

```

    name: vm-flatcar-ignition
spec:
  running: true
  template:
    spec:
      domain:
        cpu:
          cores: 1
        devices:
          disks:
            - disk: {bus: virtio}
              name: disk0
            - cdrom: {bus: sata, readonly: true}
              name: cloudinitdisk
          resources:
            requests:
              memory: 1024M
        volumes:
          - name: disk0
            persistentVolumeClaim:
              claimName: flatcar-amd64-3975
          - cloudInitConfigDrive:
              userData: |
                {"ignition":{"version":"3.3.0"},...}
              name: cloudinitdisk

```

QEMU (local testing)

```

# Download image and helper script
wget https://stable.release.flatcar-linux.net/amd64-
usr/current/flatcar_production_qemu.sh
chmod +x flatcar_production_qemu.sh
wget https://stable.release.flatcar-linux.net/amd64-
usr/current/flatcar_production_qemu_image.img

# Keep a fresh copy for re-provisioning experiments
mv flatcar_production_qemu_image.img flatcar_production_qemu_image.img.fresh
cp --reflink=auto flatcar_production_qemu_image.img.fresh
flatcar_production_qemu_image.img

# Boot with Ignition config
./flatcar_production_qemu.sh -i ignition.json
ssh -p 2222 core@127.0.0.1

```

Bare Metal

Method	Notes
ISO image	Boot and install interactively
PXE	Network boot; add <code>flatcar.first_boot=1</code> to kernel args
iPXE	Network boot with HTTP
<code>flatcar-install</code>	Run on existing Linux to install to disk

PXE/iPXE **must** set `flatcar.first_boot=1` as a kernel parameter (GRUB is not used, so the EFI flag check doesn't happen).

For bare metal metadata provisioning, use [Matchbox](#).

Customizing the Image

The `installing/customizing-the-image/` docs cover baking custom content into a Flatcar image at build time — useful for pre-populating `/etc` or adding extensions before deployment. See also `sysext-bakery`'s `bake_flatcar_image.sh`.

Migrating from CoreOS Container Linux

Flatcar is a drop-in replacement. Key naming changes:

CoreOS	Flatcar
<code>coreos-installer</code>	<code>flatcar-installer</code>
<code>coreos.first_boot=1</code>	<code>flatcar.first_boot=1</code>
<code>coreos.config.url=URL</code>	<code>ignition.config.url=URL</code>
<code>coreos.oem.id=NAME</code>	<code>flatcar.oem.id=NAME</code>
QEMU fw_cfg: <code>opt/com.coreos/config</code>	<code>opt/org.flatcar-linux/config</code>
VMware: <code>coreos.config.data</code>	<code>ignition.config.data</code>

In-place update from CoreOS CL to Flatcar is supported — see [flatcar-reference](#).